MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 5680

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode

Outline: See figure 1

Mounting position: Vertical, anode down

Weight: 5-1/4 pounds nominal

ABSOLUTE RATINGS:

Parameter:	Fl	Ef	Eb	eb	Ec	Ib	ib	Pp
Unit:	MHz	Vac	kVdc	kv	Vdc	Adc	a	w
Maximum:								(see note 2)
Pulse operation:	5.0	13.0±5%	---	17.5	-5000	---	35 (see note 1)	1200
Class C, Teleg:	5.0	13.0±5%	6.0	---	-2000	2.0	---	2500
Class C, Telep:	5.0	13.0±5%	5.0	---	-2000	1.5	---	1600
TEST CONDITIONS:	---	13.0	3.5	---	---	---	---	---

GENERAL:

Qualification - Required

5960
FSC 5960

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Method	Requirement or test	Conditions	AQL	Wsp Level	Symbol	Limits		Unit
						Min	Max	
	Quality conformance inspection, part 1							
1301	Filament current		0.65	II	If	35.0	38.0	Aac
1256	Electrode current (grid)	Ec/Ib = 0.35Adc; t = 180	0.65	II	Ic	---	15	µAdc
1261	Electrode voltage (anode)	Ec = 0; Ib = 0.2 Adc	0.65	II	Eb	1.40	1.85	kVdc
1261	Electrode voltage (anode)	Ec = -50Vdc; Ib = 0.2 Adc	0.65	II	Eb	2.50	3.20	kVdc
1316	Amplification factor	Ec/Ib = 0.2Adc	0.65	II	Mu	22	28	---
1261	Electrode voltage (grid)	Ec/Ib = 100 µAdc;Eb = 15kVdc	0.65	II	Ec	---	-850	Vdc
1231	Peak emission	ec = eb = 4000v (note 4)	0.65	II	is	35.0	---	a
---	Insulation	Pp = 300; Ib = 4 to 6 mAdc; Eb = 20kVdc (see note 3 and 4)	0.65	II	---	---	---	---
---	Power output	Eb = 5.0kVdc; Ib = 1.0Adc; Ic = 0.15Adc; Rg = 4000 ohms; F = 2 MHz	0.65	II	Po	3.25	---	kW
	Quality conformance inspection, part 2							
1331	Direct-interelectrode capacitance		---	---	Cgp	11.0	15.1	pF
			---	---	Cgf	13.0	17.0	pF
			---	---	Cpf	1.30	2.30	pF
	Quality conformance inspection, part 3							
---	Life-test provisions	Group D; Power output; t = 500 hours	---	---	---	---	---	---
---	Life-test end points	Peak emission	---	---	is	28.0	---	a

NOTES:

1. Peak instantaneous cathode current during pulse with maximum duty cycle of 0.005.
2. Minimum air flow = 75 cfm for pulse conditions and 150 cfm for CW conditions.
3. Anode potential is slowly increased to 20 kVdc at which potential the time without flashovers shall not be less than 300 seconds. A capacity of not less than 0.1 mfd shall be connected in parallel with the anode supply.

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4. These tests shall be the first to be performed at the conclusion of the holding period.

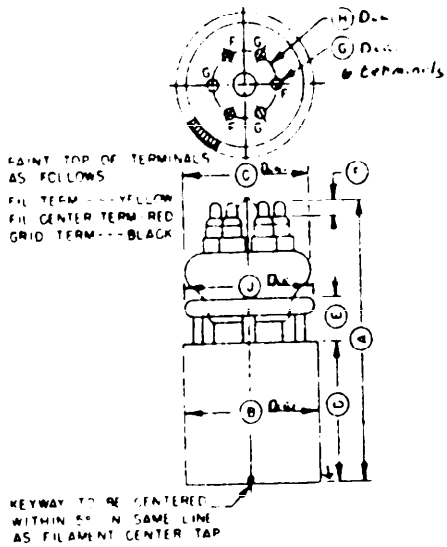
Review activity:
Navy - CG

User activities:
Navy - AS, OS, MC, SH

Preparing activity:
Navy - EC

(Project 5960 - N889)

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EC 474

Ltr	Dimensions	
	Min	Max
QC1, part 2		
A	6.50(165.1)	7.00(177.8)
B	3.469(88.11)	3.531(89.69)
D	3.375(85.73)	3.625(92.08)
QC1, part 3 (periodic-check)		
C		3.25(82.55)
E	.875(22.23)	1.000(25.40)
F	.375(9.53)	
G	.359(9.12)	.391(9.93)
H	1.375(34.93)	1.625(41.28)
J		3.750(95.25)

FIGURE 1. Outline drawing of electron tube type 5680

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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